

TADEBYAYAKHA, Russian Federation

WMO: 209640

Lat: 70.350N

Lon: 74.133E

Elev: 4

StdP: 101.28

Time Zone: 5.00 (E05)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.9	-40.4	-44.4	0.0	-42.3	-42.2	0.1	-40.0	16.5	-18.3	14.7	-17.9	2.6	70	1.079

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.6	19.6	15.0	16.6	13.6	14.0	11.8	15.4	18.8	13.7	16.2	11.9	13.9	6.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.7	9.8	16.8	12.1	8.8	15.2	10.5	7.9	13.5	43.3	19.1	38.5	16.2	34.0	14.0	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.6	13.0	11.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-10.2	-27.0	-24.9	-20.2	-17.6	-6.7	2.0	8.2	7.6	3.3	-7.3	-17.5	-23.1
	DBStd	14.36	8.48	9.19	8.97	8.36	4.97	4.15	4.79	2.86	3.18	7.07	8.96	8.76
	HDD10.0	7426	1146	978	935	829	519	246	98	86	201	535	826	1027
	HDD18.3	10409	1404	1212	1193	1079	777	491	315	332	451	793	1076	1285
	CDD10.0	59	0	0	0	0	0	6	42	12	1	0	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	12	0	0	0	0	0	1	11	1	0	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	5.6	5.9	5.8	5.9	5.6	5.8	4.8	4.5	5.2	5.5	6.2	6.0	6.2
Precipitation	PrecAvg	603	56	51	42	43	40	43	40	51	55	67	56	58
	PrecMax	778	134	135	114	80	70	80	84	98	100	96	137	101
	PrecMin	445	11	9	9	7	24	7	11	14	18	35	9	26
	PrecStd	91	26	25	21	18	12	18	21	18	20	16	21	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.0	-2.7	-0.2	1.1	3.1	19.7	24.4	19.3	13.0	5.6	0.6	-1.8
		MCWB	-6.2	-3.0	-0.4	0.5	2.4	14.0	17.2	15.7	9.8	4.7	0.0	-2.3
	2%	DB	-8.7	-5.5	-2.0	0.3	1.6	14.2	21.2	15.6	10.4	3.8	-0.5	-4.4
		MCWB	-9.0	-5.8	-2.2	-0.2	0.9	11.5	16.0	13.0	9.2	3.0	-0.7	-4.7
	5%	DB	-11.7	-7.9	-3.9	-0.8	0.9	10.1	18.9	13.7	9.0	2.6	-1.8	-7.3
		MCWB	-12.1	-8.3	-4.2	-1.3	0.3	8.3	14.5	11.7	8.1	2.0	-2.1	-7.7
	10%	DB	-14.4	-11.4	-6.6	-3.4	0.3	7.1	16.4	12.2	7.8	1.4	-3.3	-9.5
		MCWB	-14.8	-11.7	-7.0	-3.9	-0.2	5.6	13.4	10.6	6.8	0.8	-3.7	-9.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.1	-3.0	-0.4	0.6	2.6	14.5	18.4	15.7	10.5	5.0	0.2	-2.2
		MCDB	-6.1	-2.8	-0.3	1.0	3.0	18.5	22.4	18.6	12.0	5.6	0.3	-1.8
	2%	WB	-9.0	-5.7	-2.2	0.0	1.0	11.4	16.6	13.6	9.3	3.2	-0.8	-4.8
		MCDB	-8.9	-5.4	-2.1	0.2	1.3	14.4	20.8	15.0	10.3	3.7	-0.5	-4.6
	5%	WB	-11.9	-8.2	-4.2	-1.2	0.5	8.1	15.0	12.1	8.2	2.0	-2.1	-7.6
		MCDB	-11.6	-7.9	-3.9	-0.7	0.9	10.0	18.2	13.4	8.9	2.5	-1.9	-7.2
	10%	WB	-14.8	-11.7	-6.9	-3.8	-0.2	5.7	13.5	10.7	7.0	0.9	-3.7	-9.9
		MCDB	-14.4	-11.4	-6.7	-3.3	0.3	7.1	16.2	11.9	7.7	1.3	-3.3	-9.6
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.3	8.9	9.4	5.6	4.6	6.6	4.9	4.1	5.7	7.5	7.9
		MCDBR	12.2	10.1	9.9	7.7	4.0	9.9	11.2	8.3	5.6	3.9	8.0	10.1
		MCWBR	12.2	10.0	9.9	7.5	3.9	7.0	6.5	5.9	4.7	3.8	7.9	10.0
	5% WB	MCDBR	12.1	10.2	9.4	7.3	4.2	9.5	10.5	7.3	5.2	4.0	7.9	10.0
		MCWBR	12.1	10.1	9.5	7.3	4.1	7.0	7.0	5.8	4.5	4.0	7.9	10.0
Clear-Sky Solar Irradiance	taub	N/A	0.237	0.294	0.316	0.336	0.349	0.375	0.358	0.312	0.235	N/A	N/A	
	taud	N/A	1.955	2.027	2.061	2.100	2.310	2.341	2.383	2.415	2.104	N/A	N/A	
	Ebn at Noon	0	441	676	798	832	837	797	763	696	451	0	0	
	Edh at Noon	0	42	84	113	124	105	99	84	60	34	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.25	1.72	3.90	5.78	5.71	4.95	3.08	1.43	0.42	0.00	0.00	
	RadStd	0.00	0.04	0.16	0.36	0.29	0.41	0.63	0.39	0.15	0.06	0.00	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page